

Class14: RNASeq Mini Project

Laura Sun (PID: A17923552)

Table of contents

Background	1
Data Import	1
Remove zero count genes	3
DESeq Analysis	3
Data Visualization	5
Add Annotation	6
Pathway Analysis	7
KEGG	7
GO terms	10
Reactome	11
Save our results	11

Background

Here we work through a complete RNASeq analysis project. The input data comes from a knock-down experiment of a HOX gene.

Data Import

Reading the counts and metadata CSV files check on data structure

```
counts <- read.csv("GSE37704_featurecounts.csv", row.names=1)
metadata <- read.csv("GSE37704_metadata.csv")
```

```
head(counts)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

```
metadata
```

	id	condition
1	SRR493366	control_sirna
2	SRR493367	control_sirna
3	SRR493368	control_sirna
4	SRR493369	hoxa1_kd
5	SRR493370	hoxa1_kd
6	SRR493371	hoxa1_kd

Some book-keeping is required as there looks to be a mismatch between metadata rows and counts columns.

```
ncol(counts)
```

```
[1] 7
```

```
nrow(metadata)
```

```
[1] 6
```

Looks like we need to get rid of the first “length” column of our `counts` object.

```
cleancounts <- counts[,-1]
```

```
colnames(cleancounts)
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
metadata$id
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

```
all(colnames(cleancounts) == metadata$id)
```

```
[1] TRUE
```

Remove zero count genes

There are lots of genes with zero counts. We can remove these from future analysis.

```
head(cleancounts)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

```
to.keep.inds <- rowSums(cleancounts) > 0  
nonzero_counts <- cleancounts[to.keep.inds,]
```

DESeq Analysis

Load the package

```
library(DESeq2)
```

Setup DESeq object

```
dds <- DESeqDataSetFromMatrix(countData = nonzero_counts, colData = metadata, design = ~cond.
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

run DESeq

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

get results

```
res <- results(dds)
head(res)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01

padj

```

      <numeric>
ENSG00000279457 6.86555e-01
ENSG00000187634 5.15718e-03
ENSG00000188976 1.76549e-35
ENSG00000187961 1.13413e-07
ENSG00000187583 9.19031e-01
ENSG00000187642 4.03379e-01

```

Data Visualization

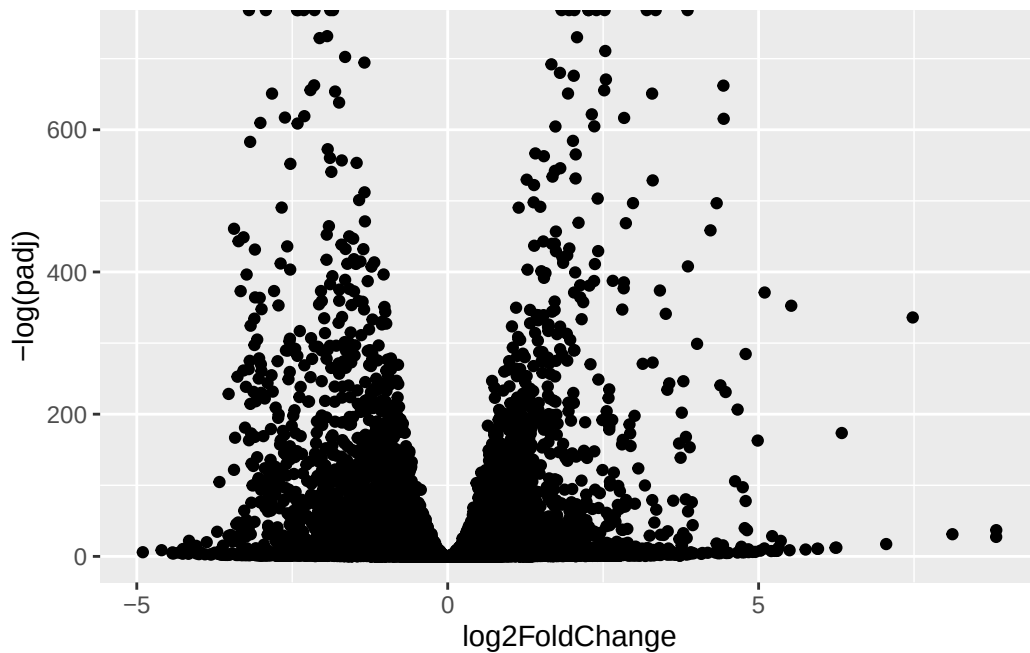
volcano plot

```

library(ggplot2)
ggplot(res) + aes(log2FoldChange, -log(padj)) + geom_point()

```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Add threshold lines for fold-change and P-value and color our subset of genes that make these threshold cut-offs in the plot.

```

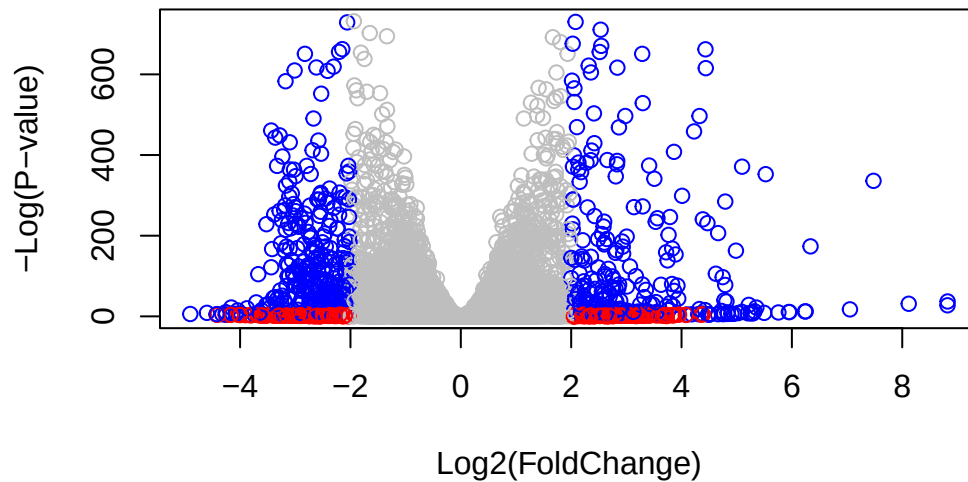
mycols <- rep("grey", nrow(res))

# Color the genes with absolute fold change above 2 to be red.
mycols[abs(res$log2FoldChange) > 2 ] <- "red"

# Color those with adjusted p-value less than 0.01 to be blue.
# and absolute fold change more than 2
inds <- (res$padj<0.01) & (abs(res$log2FoldChange) > 2 )
mycols[ inds ] <- "blue"

plot( res$log2FoldChange, -log(res$padj), col=mycols, xlab="Log2(FoldChange)", ylab="-Log(P-value)" )

```



Add Annotation

Add gene symbols and entrez

```

library(AnnotationDbi)
library(org.Hs.eg.db)

```

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="SYMBOL")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="ENTREZID")
```

'select()' returned 1:many mapping between keys and columns

Pathway Analysis

KEGG

Run gage analysis with KEGG

```
library(gage)
library(gageData)
library(pathview)
```

We need a named vector of fold-change values as input for gage.

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

	<NA>	148398	26155	339451	84069	84808
	0.17925708	0.42645712	-0.69272046	0.72975561	0.04057653	0.54281049

```
data(kegg.sets.hs)
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

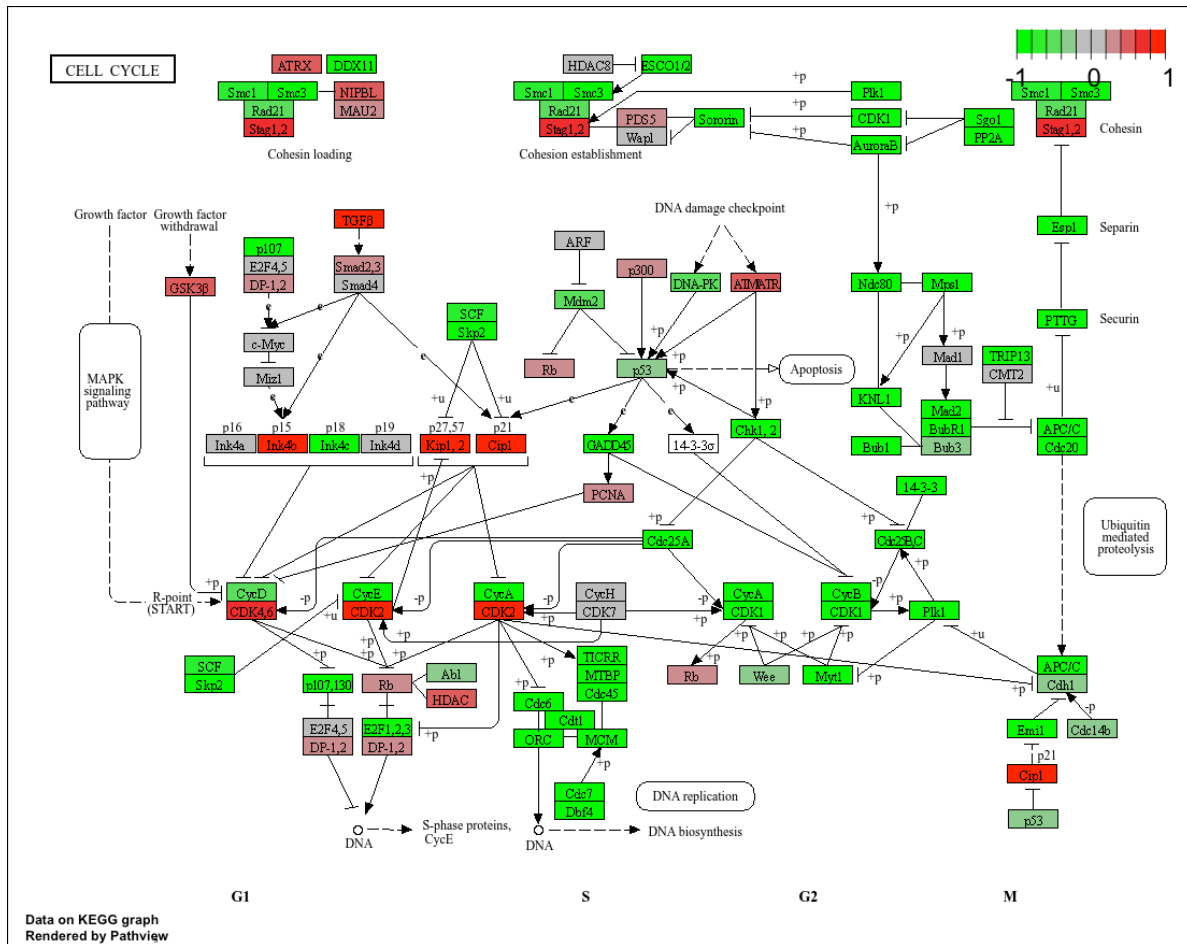
```
head(keggres$less,5)
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/laurasun/Desktop/3rd Year/BIMM143/Wk7Class14

Info: Writing image file hsa04110.pathview.png



Finding the 5 down-regulated pathways:

```
keggrespathways <- rownames(keggres$less)[1:5]
```

```
# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```



```
[1] "hsa04110" "hsa03030" "hsa05130" "hsa03013" "hsa03440"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/laurasun/Desktop/3rd Year/BIMM143/Wk7Class14
```

```
Info: Writing image file hsa04110.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/laurasun/Desktop/3rd Year/BIMM143/Wk7Class14
```

```
Info: Writing image file hsa03030.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/laurasun/Desktop/3rd Year/BIMM143/Wk7Class14
```

```
Info: Writing image file hsa05130.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/laurasun/Desktop/3rd Year/BIMM143/Wk7Class14
```

```
Info: Writing image file hsa03013.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/laurasun/Desktop/3rd Year/BIMM143/Wk7Class14
```

```
Info: Writing image file hsa03440.pathview.png
```

[illegible]

Same analysis but using GO genesets rather than KEGG.

10

```
head(gobpres$less, 4)
```

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280	nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067	mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14

		q.val	set.size	expl
G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14

Reactome

Many people like reactome web interface. You can also run this as an R function but let's look at the website first (<https://reactome.org/>)

The website wants a text file with one gene symbol per line of the genes you want to map to pathways.

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
head(sig_genes) #res$symbol
```

```
ENSG000000187634 ENSG000000188976 ENSG000000187961 ENSG000000188290 ENSG000000187608
      "SAMD11"      "NOC2L"      "KLHL17"      "HES4"      "ISG15"
ENSG000000188157
      "AGRN"
```

and write out to a file

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

In the event hierarchy in Reactome website, “autophagy” is on the top. I saw the text “Entities pvalue” but there's no number under it.

Save our results

```
write.csv(res,file="myresults.csv")
```